

Bridging Graph Polynomials and Symmetric Tensors

TENORS Workshop 2

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 - 2 Attempting a Belkale-Brosnan type result for symmetric tensors

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- Notation: $V^{\otimes d}$ denotes the tensor product of a vector space V with itself d times. $S^d(V) \subset V^{\otimes d}$ denotes the subspace containing all the symmetric tensors.
- Elementary symmetric tensors are of the form $v^{\otimes d}$.

Relationship with Homogenous Polynomials

Consider $k[x_1 \dots x_n]_d$, the set of all homogenous polynomials of degree d over k (implicitly included in this is the 0 polynomial). Then an established result [3] is that

$$k[x_1 \dots x_n]_d \cong S^d(k^n)$$

as k -vector spaces, by linearly extending the mapping of monomials

$$\mathbf{x}^\alpha := \prod_{i=1}^n x_i^{a_i} \mapsto \frac{1}{d!} \sum_{\sigma \in S_d} \sigma \left(\bigotimes_{i=1}^n e_i^{\otimes a_i} \right)$$

Here α denotes the tuple $(a_1, a_2, \dots, a_n)$ where $a_i \geq 0 \forall i = 1, 2, \dots, n$, and for this mapping $|\alpha| = \sum_{i=1}^n a_i = d$.

Veronese Variety

Geometrically, consider $\mathbb{P}V$ to be the set of all lines in $V \cong k^n$ passing through the origin. Then the Veronese mapping given by

$$\nu_d : \mathbb{P}V \rightarrow \mathbb{P}k^N$$

taking points

$$[x_1 : x_2 : \dots : x_n] \mapsto [x_1^d : x_1^{d-1}x_2 : \dots : x_n^d]$$

allows for the visualisation of elementary (rank-1) symmetric tensors as points on a variety in $\mathbb{P}k^N$. Here $N = \binom{n-1+d}{d}$ [9]. More general symmetric tensors then lie on secant varieties to the Veronese.

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- A *tree* is a connected graph with no loops i.e. $b_0(G) = 1, b_1(G) = 0$.
- A *spanning* tree $T \leq G$ contains all of V but not (necessarily) all of E .

Constructing (particular) Graph Polynomials

For the rest of this presentation $G = (V, E)$ such that $\|G\| = n$ and $b_0(G) = 1$. A *Kirchoff* polynomial [5, 10] for a given graph G is given by

$$P_G(x_1, \dots, x_n) := \sum_{T \leq G} \prod_{e \notin T} x_e$$

Note the variables of the polynomial correspond to a choice of labelling the edges.

Properties of the Kirchoff Polynomial

- They are homogenous polynomials, and lie in the quotient ring $R := k[x_1, x_2, \dots, x_n]/(x_1^2, x_2^2, \dots, x_n^2)$

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- Variables common to each term in the polynomial denote self-loops of G .
- An equivalence relation $\sim_S$ can be imposed by $P_A \sim_S P_B \iff P_A(x_1, \dots, x_n) = \tau \cdot P_B(x_1, \dots, x_n)$. Here $\tau \in S_n$ permutes n elements, corresponding to a relabelling of edges.

What is... the Grothendieck Ring of Varieties?

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 $[X] = \{V | \pi : X \dashrightarrow V \text{ a birational morphism exists}\}$; modulo all relations of the form $[X] = [X \setminus Y] + [Y]$.
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- This group can be seen as a commutative ring with unity by defining multiplication $[X][Y] = [X \times_k Y]$.
- $\mathcal{K}(V_k)$ can be seen as a module over $\mathbb{Z}[\mathbb{L}]$ where $\mathbb{L} := [\mathbb{A}^1]$ [1].

Belkale, Brosnan (2003)

Let $\mathbb{V}(P_G) \subset \mathbb{A}^n$ denote the scheme of zeroes of P_G . Define $Y_G = \mathbb{A}^n \setminus \mathbb{V}(P_G)$.

Theorem (Belkale, Brosnan (2003) [2])

Define $S := \{\mathbb{L}^n - \mathbb{L}\}_{n \geq 2}$ as a multiplicative subset in $\mathcal{K}(V_k)$. Then the collection of $[Y_G]$ generate $S^{-1}\mathcal{K}(V_k)$ as a $S^{-1}\mathbb{Z}[\mathbb{L}]$ module.

Aluffi, Marcolli 2011

X, Y are stably birational $\iff X \times \mathbb{P}k^m \dashrightarrow Y \times \mathbb{P}k^n$ is birational, for $n, m \geq 0$.

Theorem (Larsen, Lunts (2001) [7])

$\mathcal{K}(V_k)/(\mathbb{L}) \cong \mathbb{Z}[SB]$, the ring of stably birational varieties

Theorem (Aluffi, Marcolli (2011) [1])

The images of $[Y_G]$ in $\mathcal{K}(V_k)$ generate $\mathbb{Z} \subsetneq \mathbb{Z}[SB]$

Corollary

Suppose $S' := \{\mathbb{L}^n - 1\}_{n \geq 1}$ Then $[Y_G]$ do not generate $S'^{-1}\mathcal{K}(V_k)$ over $S'^{-1}\mathbb{Z}[\mathbb{L}]$

Graphs as a Disjoint Union

Let $\mathcal{G}_n$ denote the family of all connected graphs with a fixed number of edges n .

- The corresponding graph polynomials all lie in $R := k[x_1, \dots, x_n]/(x_1^2, \dots, x_n^2)$.

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- Now $\mathbf{G}_n := (\mathcal{G}_n / \sim_{KP}) / \sim_S$ partitions the space of finite graphs into classes having the same Kirchoff polynomial upto relabelling.
- $\mathbf{G}_n$ can then be expressed as the disjoint union $\bigsqcup_{i=0}^n \mathbf{G}_{n,i}$ where each component contains the graphs with $b_1(G) = i$.

Addition in Graphs?

In $\mathbf{G}_n$, an attempt to define an addition operation is given below

- Let $[A]_{KP,S}, [B]_{KP,S} \in \mathbf{G}_n := (\mathcal{G}_n / \sim_{KP}) / \sim_S$

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- The sum $[P_A]_S + [P_B]_S$ as a set can be seen as $\{p(x_1, \dots, x_n) + q(x_1, \dots, x_n) | \sigma \cdot p = P_A, \tau \cdot q = P_B, \sigma, \tau \in S_n\}$

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- P_C can be chosen as an element with the largest number of summands in this set. $[P_C]_S$ then is a corresponding equivalence class.
- **Conjecture:** Does there always exist C , a graph, such that its Kirchhoff polynomial is P_C ? If yes then
- *This turns $\mathbf{G}_n$ into an additive abelian group*

Image of R in symmetric tensors

What does R (the ring of Kirchoff polynomials) look like when using Φ the isomorphism between homogenous polynomials and symmetric tensors?

(**Note:** k is no longer required to be $\mathbb{F}_2$)

- Definitely consists of tensors that are not elementary, because R contains no monomials with exponents larger than 1.

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- Since R describes a quotient ring of homogenous polynomials, it must also induce images in the space of symmetric tensors
- However, since none of the elements in R can be expressed as linear polynomials to the d -th power, they cannot be seen as lying on the Veronese variety.
- Consider the mapping $\mu_d : \mathbb{P}V \rightarrow \mathbb{P}k^{\binom{n}{d}}$ taking $[x_1 : \dots : x_n] \mapsto [\prod_{i=1}^d x_i : \dots : \prod_{i=n-d+1}^n x_i]$.

Kirchoff Polynomials as Symmetric Tensors

$$\begin{array}{ccc}
 \mathbb{P}V & \xrightarrow{\nu_d} & \mathbb{P}k^{\binom{n+d-1}{d}} \\
 & \searrow \mu_d & \nearrow i \\
 & \mathbb{P}k^{\binom{n}{d}} &
 \end{array}$$

Thus Kirchoff polynomials can be seen as specific points on $\sigma_r(i(\mu_d(\mathbb{P}V))) \subseteq \sigma_r(\nu_d(\mathbb{P}V))$ where r denotes the rank of the image tensor under Φ .

Symmetric Tensors and the Grothendieck Ring

If graph polynomials can generate the Grothendieck Ring (in a localised manner), can the equivalent statement be transported to working with symmetric tensors?

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- Every symmetric tensor can be associated therefore to a variety (and its complement)
- In particular the symmetric tensors associated to graph polynomials then correspond to $[Y_G]$ as defined earlier as well as the complement $\mathbb{V}(P_G)$.

Generation of Symmetric Tensors

What does the element $[\mathbb{A}^1] =: \mathbb{L} \in \mathcal{K}(V_k)$ correspond to when working with symmetric tensors or homogenous polynomials?

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- In n variables, the intersection of $(n - 1)$ of the $\mathbb{V}(x_i)$ is birational to $\mathbb{A}^1$
- Describing the intersection in terms of a single homogenous polynomials, or in terms of an ideal generated by these linear monomials may be a profitable next step.

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- What is the significance of the image $\mathbb{L}$ when working with symmetric tensors?
- Do questions about rank/rank decompositions get answered by this approach?
- Does applying a grading on the space of graphs only work when dealing with the $\mathbb{F}_2$? Can this be extended to any field k ?

Bibliography I

- [1] Paolo Aluffi and Matilde Marcolli, *Graph hypersurfaces and a dichotomy in the Grothendieck ring*, Letters in Mathematical Physics **95** (March 2011), no. 3, 223–232 (en). arXiv:1005.4470 [math].
- [2] Prakash Belkale and Patrick Brosnan, *Matroids motives, and a conjecture of Kontsevich*, Duke Mathematical Journal **116** (January 2003), no. 1 (en).
- [3] Pierre Comon, Gene Golub, Lek-Heng Lim, and Bernard Mourrain, *Symmetric Tensors and Symmetric Tensor Rank*, SIAM Journal on Matrix Analysis and Applications **30** (January 2008), no. 3, 1254–1279 (en).
- [4] Reinhard Diestel, *Graph Theory*, Graduate Texts in Mathematics, vol. 173, Springer Berlin Heidelberg, Berlin, Heidelberg, 2017 (en).
- [5] Thomas Krajewski, Vincent Rivasseau, Adrian Tanasă, and Zhituo Wang, *Topological graph polynomials and quantum field theory Part I: heat kernel theories*, Journal of Noncommutative Geometry **4** (January 2010), no. 1, 29–82 (en).
- [6] Joseph M. Landsberg, *Tensors: geometry and applications*, Graduate studies in mathematics, American mathematical society, Providence (R.I.), 2012 (eng).

Bibliography II

- [7] Michael Larsen and Valery A Lunts, *Motivic measures and stable birational geometry*, arXiv preprint math (2001).
- [8] Bjorn Poonen, *The Grothendieck ring of varieties is not a domain*, Mathematical Research Letters **9** (2002), no. 4, 493–497 (en). arXiv:math/0204306.
- [9] Karen E. Smith (ed.), *An invitation to algebraic geometry*, Corrected printing, Universitext, Springer, New York Berlin Heidelberg, 2004 (en).
- [10] Richard P. Stanley, *Spanning trees and a conjecture of Kontsevich*, Annals of Combinatorics **2** (December 1998), no. 4, 351–363 (en).